

Joint Feature Selection and Classifier Design for Radar Targets

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Abstract

A joint feature selection and classifier design is proposed in this paper. The approach adopts the feature dimension extension by power transformation as a new kernel function, which can not only make full use of the input samples to form the nonlinear classification boundary, but also realize the nonlinear feature selection. A zero-mean Gaussian prior with Gamma precision is used to promote sparsity in utilization of features in our model. The experiments based on a measured radar data set demonstrate the practicability and effectiveness of the proposed method.

Keywords: classifier design, feature selection, sparsity, Relevance Vector Machine (RVM), variational bayesian (VB) inference.

1. Introduction

In binary supervised learning problems, a data set of input-label pairs $\{\mathbf{x}_n, z_n\}_{n=1}^N$ is given. Usually, the input sample is a p -dimensional feature vector $\mathbf{x}_n = [x_{1n}, x_{2n}, \dots, x_{pn}] \in \mathbb{R}^p$, and the corresponding label is $z_n \in \{1, -1\}$, we wish to learn a function to describe the dependence of the labels on the inputs with the objective of making accurate predictions of z_* for a new sample $\mathbf{x}_*$ (known as classifier design), and to identify a subset of features from the high-dimensional feature vector that is most informative about the class distinction (known as feature selection). In this paper, we develop a *joint feature selection and classifier design* (referred to as JFSCD) to accomplish both classifier design and feature selection simultaneously.

Tipping proposed the *Relevance Vector Machine* (RVM) [1], a probabilistic model of identical functional form to the *Support Vector Machine* (SVM) [2], RVM achieves comparable accuracy and relative sparsity to SVM, yet provides a full predictive distribution [1]. Nevertheless, both of them do not have the ability to select features.

Since RVM is highly sparse, *Relevance Vector Machine Feature Selection and Classification for Underwater Targets* (RVMFSC) is proposed in [3] to realize not only classification but also linear feature selection by imposing sparsity in feature space, where data is originally represented, via the linear kernel. Nevertheless, RVMFSC can just construct the linear classification boundary, thereby, it cannot guarantee the good performances for some nonlinear classification problems.

This paper seeks an alternative Bayesian approach to realize both nonlinear classification and nonlinear feature

selection via the feature dimension extension by power transformation, in which features are put on sparsity-promoting prior.

2. A joint feature selection and classifier design

2.1. Basic model construction

Although the most common link between regression and classification is the Logistic function, here we adopt the Probit link [4]. The fundamental reasons behind our choice of the Probit link are its simple interpretation in terms of hidden variable and its general extension to multiclass classifier via multinomial Probit. The basic construction of our model is as follows:

$$y_n = \tilde{\mathbf{x}}_n \mathbf{w} + \varepsilon_n, \quad \varepsilon_n \sim N(0, \tau^{-1}),$$

$$p(z_n | y_n) = I(z_n = \text{sign}(y_n)) = \begin{cases} 1 & z_n = \text{sign}(y_n) \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

where the notation $N(\cdot)$ represents the Gaussian distribution, $I(\cdot)$ is an indicator function, and y_n is a continuous random variable hidden behind the label z_n . In Eq. (1), $\tilde{\mathbf{x}}_n = [\mathbf{x}_n^1, \dots, \mathbf{x}_n^m, \dots, \mathbf{x}_n^M]$ is the power transformation extension of the input sample $\mathbf{x}_n$, where $m \in \{1, \dots, M\}$ is the power parameter, and M (usually $M \geq 2$) is the maximum power. The corresponding weight is also extended as $\mathbf{w} = [\tilde{\mathbf{w}}_1, \dots, \tilde{\mathbf{w}}_m, \dots, \tilde{\mathbf{w}}_M]^T$ with $\tilde{\mathbf{w}}_m = [w_{1m}, w_{2m}, \dots, w_{pm}]$.

In Eq. (1), ε_n is the independent noise [1] which is assumed to be zero-mean Gaussian with variance τ^{-1} . Thus $p(y_n | \mathbf{x}_n) = N(y_n | \tilde{\mathbf{x}}_n \mathbf{w}, \tau^{-1})$, and the model for binary classification is:

$$p(z_n, y_n | \tilde{\mathbf{x}}_n \mathbf{w}, \tau) = p(z_n | y_n) N(y_n; \tilde{\mathbf{x}}_n \mathbf{w}, \tau^{-1}) \quad (2)$$

By integrating out y_n , we can obtain the Probit link :

$$p(z_n = 1 | \tilde{\mathbf{x}}_n \mathbf{w}, \tau) = \int p(z_n = 1, y_n | \tilde{\mathbf{x}}_n \mathbf{w}, \tau) dy_n$$

$$= \int_0^\infty N(y_n; \tilde{\mathbf{x}}_n \mathbf{w}, \tau^{-1}) dy_n \quad (3)$$

$$= \text{normcdf}\left(\frac{\tilde{\mathbf{x}}_n \mathbf{w}}{\tau^{-1/2}}\right)$$

where $\text{normcdf}(\cdot)$ denotes the Normal cumulative distribution function.

2.2 . Sparsity-promoting prior

We adopt a zero-mean Gaussian prior [1] to favor a feature-sparse estimate for $\mathbf{w}$, that is:

$$p(\mathbf{w} | \boldsymbol{\alpha}) = \prod_{p=1}^P \prod_{m=1}^M N(\mathbf{w}_{pm} | 0, \alpha_p^{-1}) \quad (4)$$

where $\boldsymbol{\alpha} = [\alpha_1, \dots, \alpha_M]^T$ with $\tilde{\boldsymbol{\alpha}}_m = [\alpha_1, \alpha_2, \dots, \alpha_p]$ for $m=1, \dots, M$. To complete the specification of this hierarchical prior, we place Gamma priors [1] on $\boldsymbol{\alpha}$ and the noise precision τ :

$$p(\boldsymbol{\alpha}) = \prod_{m=1}^M \prod_{p=1}^P \text{Gamma}(\alpha_p | a, b) = \prod_{m=1}^M \prod_{p=1}^P \Gamma(a)^{-1} b^a \alpha_p^{a-1} e^{-b\alpha_p} \quad (5)$$

$$p(\tau) = \text{Gamma}(\tau | c, d)$$

with $\text{Gamma}(\cdot)$ denoting the Gamma distribution, and $\Gamma(\cdot)$ representing the gamma function. It has been demonstrated [1] that appropriate choice of parameters a and b encourages a sparse representation for α_p . Typically, when $a = b = \varsigma$, with $\varsigma > 0$ a small constant (e.g., $\varsigma = 10^{-6}$ in the following experiment), $\text{Gamma}(\alpha_p | a, b)$ has a large spike concentrated at zero and a heavy right tail. The spike corresponds to the most discriminative or informative features. At the other extreme, features for which α_p is in the right tail have $\{\mathbf{w}_{pm}\}_{m=1}^M$ that are shrunk strongly to zero, favoring sparseness.

The above hierarchical prior is similar to that in RVM, but the difference is that here $\boldsymbol{\alpha}$ is also extended with the power transformations. Notice that, here $\{\mathbf{w}_{pm}\}_{m=1}^M$ for each p can vary across the M power transformations, which means that different powers of the same feature can play different roles in constructing the classification boundary, however, we expect α_p is set to be fixed for different powers of the p th feature. i.e., $\tilde{\boldsymbol{\alpha}}_m$ are shared among all M power transformed data, which means that each feature with all M power transformations $\{\mathbf{x}_{pm}^m\}_{m=1}^M$ will contribute to learning the α_p , offering the opportunity to be selected or not selected for all $m=1, \dots, M$. Therefore, the proposed prior structure can promote the sparse feature selection, and form the nonlinear classification boundary as well.

After introducing the model, we now have a complete probabilistic specification of this model, with a graphical model representation illustrated in Fig.1.

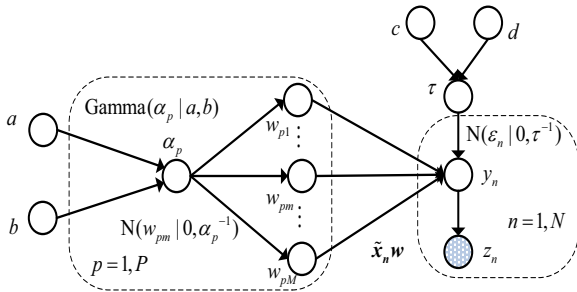


Fig. 1. Graphical representation of the proposed JFSCD model.

3. Variational bayesian inference

Bayesian inference seeks to estimate the posterior distribution of the latent variables Ψ , given the observed data $\mathbf{D}$ and hyper-parameters $\boldsymbol{\theta}$. We employ variational Bayesian (VB) inference [5] as a compromise between accuracy and efficiency. This method approximates an intractable joint posterior $p(\Psi | \mathbf{D})$ of all the hidden variables by a product of marginal distributions $q(\Psi) = \prod_f q(\Psi_f)$, each over only a single latent variable Ψ_f . The optimal parameterization of $q(\Psi_f)$ is obtained by minimizing the Kullback-Leibler divergence between the variational approximation $q(\Psi)$ and the true joint posterior $p(\Psi | \mathbf{D}, \boldsymbol{\theta})$.

In the JFSCD model, $\mathbf{D}$ is $\{\mathbf{x}_n, \mathbf{z}_n\}_{n=1}^N$, Ψ is $\{y_n, \mathbf{w}_{pm}, \alpha_p, \tau\}_{n=1}^N, \{m=1, p=1\}^{M \times P}$, and $\boldsymbol{\theta}$ is $\{a, b, c, d\}$. Due to the conjugacy properties of the $\{p(\Psi_f | \mathbf{D}, \boldsymbol{\theta})\}$, we can evaluate $\{q(\Psi_f)\}$ analytically. The VB-based Bayesian inference for the model is summarized in Algorithm 1, which starts from a set of initial values $\boldsymbol{\theta}_0$, *maxit* represents the maximum number of the iterations, and ς is a small positive constant (we set $\varsigma = 10^{-6}$).

Algorithm 1: VB-based Bayesian Inference

$[\Psi, \boldsymbol{\theta}, F] = \text{JFSCDvb}(\boldsymbol{\theta}_0, \text{maxit}, \varsigma)$

$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}_0$

$$F(1) = \int q(\Psi) \ln \frac{p(\mathbf{D}, \Psi)}{q(\Psi)} d\Psi$$

iter = 0; *change* = 1

While (*iter* < *maxit*) && (*change* >= ς)

iter = *iter* + 1

$$q(y_n) = N^{\tau_n}(\tilde{m}_n, \tilde{\sigma}_n), \tilde{m}_n = \tilde{\mathbf{x}}_n^T \langle \mathbf{w} \rangle, \tilde{\sigma}_n = \langle \tau \rangle^{-1}$$

% $N^{\tau_n}(\cdot)$ denotes a truncated normal distribution [6] with the direction of truncation ($N^+(\cdot)$ or $N^-(\cdot)$) determined by z_n , and $\langle \cdot \rangle$ denotes the posterior expectation for the variable over the corresponding distribution on it.

$$q(\mathbf{w}) = N(\tilde{\mathbf{w}}, \tilde{\Sigma}_w)$$

$$\tilde{\Sigma}_w = (\text{diag}(\langle \alpha_p \rangle) \sum_{n=1}^N \tilde{\mathbf{x}}_n^T \tilde{\mathbf{x}}_n)^{-1} \quad \tilde{\mathbf{w}} = \langle \tau \rangle \tilde{\Sigma}_w \sum_{n=1}^N \tilde{\mathbf{x}}_n^T \langle y_n \rangle$$

$$q(\alpha_p) = \text{Gamma}(\tilde{a}_p, \tilde{b}_p), \tilde{a}_p = a_p + \frac{1}{2}, \tilde{b}_p = b_p + \frac{1}{M} \sum_{m=1}^M \frac{1}{2} \langle \mathbf{w}_{pm}^2 \rangle$$

$$q(\tau) = \text{Gamma}(\tilde{c}, \tilde{d})$$

$$\tilde{c} = c + \frac{1}{2}, \tilde{d} = d + \frac{1}{2N} \sum_{n=1}^N (\langle y_n^2 \rangle - 2 \langle y_n \rangle \tilde{\mathbf{x}}_n^T \langle \mathbf{w} \rangle + \tilde{\mathbf{x}}_n^T \langle \mathbf{w} \mathbf{w}^T \rangle \tilde{\mathbf{x}}_n^T)$$

$$F(\text{iter} + 1) = \int q(\Psi) \ln \frac{p(\mathbf{D}, \Psi)}{q(\Psi)} d\Psi$$

$$\text{change} = (F(\text{iter} + 1) - F(\text{iter})) / F(\text{iter})$$

End

After the model is trained, for a testing data sample $\mathbf{x}_*$, the predictive distribution can be calculated as:

$$\begin{aligned}
p(z_* = 1 | \mathbf{x}_*) &= \int p(z_* = 1 | y_*) p(y_* | \mathbf{x}_*, \langle \mathbf{w} \rangle, \langle \tau \rangle) dy_* \\
&= \int_0^1 N(y_*; \tilde{\mathbf{x}}_* \langle \mathbf{w} \rangle, \langle \tau \rangle^{-1}) dy_* \\
&= \text{normcdf}\left(\frac{\tilde{\mathbf{x}}_* \langle \mathbf{w} \rangle}{\langle \tau \rangle^{-1/2}}\right)
\end{aligned} \quad (6)$$

4. Experimental results

In this section we present experimental results on the measured radar data set. To obtain the meaningful comparisons with other algorithms, we use the same training samples and test samples in the experiments, and all samples are initially normalized to have zero mean and unit variance. The measured radar data set includes two classes: the car, containing 553 samples and the airplane, having 546 samples, and each sample has 21 features. The data set is averagely divided into five groups. Firstly, we randomly choose one group and fix it as the test samples; then, we decide a group as the training sample which can be chosen from the remaining four groups; finally, the other three groups are utilized as cross-validation samples in the training phase. The goal of cross-validation is to choose the parameters, if we choose several parameters at the same time, the results are averaged. As discussed above, there are four experiments with the different selection of training samples. The four experimental results are shown in Table 1, where JFSCD is compared against RVMFSC, DFDE (*direct feature dimension extension by power transformation*), RVM, RELIEF [7] joints RVM, FISHER [8] joints RVM, SVM, and exhaustive search method [8] joints SVM, of which DFDE is similar to JFSCD, but without fixing α_p for different powers of the P th feature (here $p = 1, 2, \dots, 21$), by such model structure, DFDE does not possess the ability to select features. In addition, to make accurate comparison with JFSCD, here, the number of features selected by RELIEF, FISHER and exhaustive search method is set to be equal to that of JFSCD. The experimental results includes the selected features, the error rate and the *area under curve* (AUC) [9], where AUC is another important criterion to evaluate the performance of the classifier except for classification rate, and its values belong to $[0, 1]$, the larger the value is, the better the performance is.

Table 1: From (a) to (d) respectively shows the four experimental results of the comparison between the method we proposed and other methods.

Method	Number of selected features (the corresponding labels)	Error rate	AUC
JFSCD	3(2,9,6)	1.92%	0.9975
DFDE	\	2.73%	0.9979
RVMFSC	4(6,2,3,9)	2.73%	0.9908
RVM	\	4.55%	0.9884
SVM	\	1.82%	0.9920
FISHER+RVM	3(15,3,2)	5.45%	0.9778
RELIEF+RVM	3(9,2,15)	6.21%	0.9760
Exhaustive search +SVM	3(\)	2.27%	0.9955

(a)

Method	Number of selected features (the corresponding labels)	Error rate	AUC
JFSCD	4(2,9,6,8)	1.82%	0.9979
DFDE	\	2.27%	0.9888
RVMFSC	5(2,6,9,10,21)	2.27%	0.9937
RVM	\	2.50%	0.9912
SVM	\	3.18%	0.9924
FISHER+RVM	4(15,2,16,6)	1.82%	0.9973
RELIEF+RVM	4(9,2,15,11)	4.55%	0.9805
Exhaustive search +SVM	4(2,9,10,20)	2.27%	0.9964

(b)

Method	Number of selected features (the corresponding labels)	Error rate	AUC
JFSCD	3(2,9,6)	1.82%	0.9974
DFDE	\	1.82%	0.9987
RVMFSC	4(2,6,9,10)	3.18%	0.9907
RVM	\	2.96%	0.9907
SVM	\	4.32%	0.9893
FISHER+RVM	3(15,2,3)	5.45%	0.9801
RELIEF+RVM	3(9,2,15)	5.45%	0.9757
Exhaustive search +SVM	3(\)	2.44%	0.9958

(c)

Method	Number of selected features (the corresponding labels)	Error rate	AUC
JFSCD	4(2,9,8,6)	1.82%	0.9975
DFDE	\	2.73%	0.9935
RVMFSC	5(6,2,3,18,9)	4.09%	0.9931
RVM	\	3.79%	0.9884
SVM	\	2.05%	0.9900
FISHER+RVM	4(15,3,2,16)	5.91%	0.9811
RELIEF+RVM	4(9,2,15,11)	5.18%	0.9769
Exhaustive search +SVM	4(\)	2.73%	0.9929

(d)

In Table 1, the boldface represents the best result of all methods, and the symbol “\” represents that the method cannot select features. In addition, the order of the selected features is decided by their contribution to the classification (in experiments (a), (c) and (d), exhaustive search method joints SVM does not give the selected features, that is because several feature combinations are chosen in cross-validation, which corresponds to not selecting robust feature combination). From Table 1, we can see that the classification rate and AUC obtained by our method are either the best or very close the best, and the features selected by JFSCD is steady comparing with other methods in the four experiments.

To compare these methods completely, Table 2 gives the mean and standard deviation of the error rate and AUC, respectively. The boldface also represents the best result of all methods. From Table 2, we can see that JFSCD outperforms all the other methods of all the results, which indicates that our method has good and robust performance in classification and AUC.

Table 2: the mean and standard deviation of error rate and AUC in the four experiments.

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Method	Mean error rate (/standard deviation)	Mean AUC (/standard deviation)
JFSCD	1.845%/(0.0433)	0.9976/(1.9203e-004)
DFDE	2.388%/(0.3776)	0.9947/(0.0040)
RVMFSC	3.068%/(0.6723)	0.9921/(0.0013)
RVM	3.450%/(0.7855)	0.9897/(0.0013)
SVM	2.843%/(0.9963)	0.9909/(0.0013)
FISHER+RVM	4.658%/(1.6490)	0.9841/(0.0077)
RELIEF+RVM	5.348%/(0.5955)	0.9773/(0.0019)
Exhaustive search +SVM	2.428%/(0.1879)	0.9952/(0.0013)

5. Conclusions

A JFSCD algorithm which joints classification and feature selection together via using the feature dimension extension by power transformation as a new kernel function is proposed in this paper. Experimental results indicate that the classification accuracy, AUC and feature selection performance of JFSCD are likely to be statistically superior to other methods. And our method can not only be used for radar data sets, but also is adaptive for other data sets.

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